

Memo 05: Spatial Distribution Checking for FVCOM-MP Cases a, b, and c

FVCOM-MP Delaware Test Cases

May 2026

Purpose

Memo 04 confirmed that the in-run mass-conservation gaps between the baseline case **a** and flocc-enabled cases **b/c** are resolved by Fix B+ (`isnan()` guards on erosion/deposition fluxes) and that `adv_scal` is safe. The next diagnostic step is to verify that the scalar concentration fields — microplastic (`mp1`) and suspended sediment (`coarse_sand_1`) — are spatially reasonable over the first ten days of simulation: no localized accumulation, no runaway concentrations, and physically plausible gradients across the Delaware River and Bay domain.

This memo documents the extraction methodology, the spatial distributions observed, and a comparison of the three cases at eight monitoring stations.

Objective

- (1) Verify the absence of localized numerical accumulation in the `mp1` water-column and bed-mass fields.
- (2) Compare surface-layer, bottom-layer, and depth-averaged `mp1` concentrations among cases **a**, **b**, and **c**.
- (3) Examine the flocc-component split (`mp1_agg`, `mp1_dis`) spatial patterns in cases **b/c**.
- (4) Check suspended sediment (`coarse_sand_1`) distribution in cases **b/c**.
- (5) Extract full time series at eight representative monitoring stations along the Delaware River and Bay.

Extraction Methodology

MATLAB Extraction Scripts

Two MATLAB scripts were written following the conventions established in `extract_mp1_mass_budget_a.m` (Memo 04):

`MATLAB/extract_spatial_snapshots_a.m` Reads `OUTPUT_a/backup_data/waterPACT_a.0001.nc` and extracts the last 24 hourly output records.

`MATLAB/extract_spatial_snapshots_b.c.m` Reads `OUTPUT_b/backup_data/waterPACT_b_0001.nc` and `OUTPUT_c/backup_data/waterPACT_c_0001.nc`. In addition to the `mp1` fields, this script extracts the flocculation-component variables (`mp1_agg`, `mp1_dis`) and the suspended sediment variable (`coarse_sand_1`) where available.

Both scripts save compact v7.3 HDF5 MAT files (`spatial_snapshots_{a,b,c}.mat`) containing:

- (i) Grid metadata: `lon`, `lat`, `h`, `art1`, `nv`, $\Delta\sigma$ layer fractions.
- (ii) Spatial snapshot arrays (`node` $\times$ n_{snap}) for the last 24 hourly records:
 - Surface layer ($k = 1$), bottom layer ($k = 30$), and sigma-weighted depth-averaged concentrations.
 - Bed plastic mass density (`bot_mass_mp1`) and sediment bed fraction (`coarse_sand_1_bedfrac`).
- (iii) Full-run time-series profiles (`station` $\times$ `sigma layer` $\times$ `time`) at the eight monitoring stations listed in Table 1.

Station Locations

Table 1: Monitoring station coordinates and nearest model node used for time-series extraction.

Station	Latitude (°N)	Longitude (°E)	Notes
Chesapeake Canal	39.5350	−75.7867	C&D Canal confluence
Main stream 1	39.5644	−75.5417	Lower bay, south
Main stream 2	39.7826	−75.4562	Mid bay
Main stream 3	39.8788	−75.1951	Mid bay, east
Main stream 4	40.0147	−75.0386	Upper bay
Main stream 5	40.1244	−74.8056	Near-coast transition
Upstream Delaware	40.1956	−74.7594	Delaware River upstream
Upstream Schuylkill	39.9256	−75.2121	Schuylkill tributary

Nearest-node lookup used the haversine formula; maximum reported nearest-node distance is checked to confirm all stations map to physically reasonable locations (target: < 5 km).

Depth-Averaging Formula

The sigma-weighted depth-average is computed as:

$$\langle C \rangle_i = \frac{\sum_{k=1}^{K_b} C_{i,k} A_i D_i \Delta\sigma_{i,k}}{\sum_{k=1}^{K_b} A_i D_i \Delta\sigma_{i,k}} = \frac{\sum_{k=1}^{K_b} C_{i,k} \Delta\sigma_{i,k}}{\sum_{k=1}^{K_b} \Delta\sigma_{i,k}}$$

where $D_i = \max(h_i + \zeta_i, 0)$ is the total water depth, $\Delta\sigma_{i,k} = |\text{siglev}_{i,k} - \text{siglev}_{i,k+1}|$, and the denominator equals unity by construction. Dry nodes ($D_i = 0$) are assigned zero.

Python Analysis Script

The Python script `PYTHON/analyze_spatial_distribution.py` loads the MAT files via `h5py` and generates figures using `matplotlib.tri.Triangulation` with Gouraud shading. Four figure groups are produced:

- (1) **Day-mean plan-view maps:** for each available scalar field, the temporal mean over the last 24-hour window at each node.
- (2) **Pointwise-max maps:** the temporal maximum over the last 24-hour window at each node (identifies worst-case accumulation spots).
- (3) **Cross-case difference maps:** $\text{day-mean(b)} - \text{day-mean(a)}$, $\text{day-mean(c)} - \text{day-mean(a)}$, and $\text{day-mean(c)} - \text{day-mean(b)}$ for `mp1_surf`, `mp1_davg`, and `bot_mass_mp1`.
- (4) **Station time-series plots:** surface-layer and depth-averaged mp1, and (for cases b/c) `coarse_sand_1` at all eight stations.

All figures are saved to `PYTHON/output/spatial_distribution/`.

Results

Day-Mean Spatial Distributions

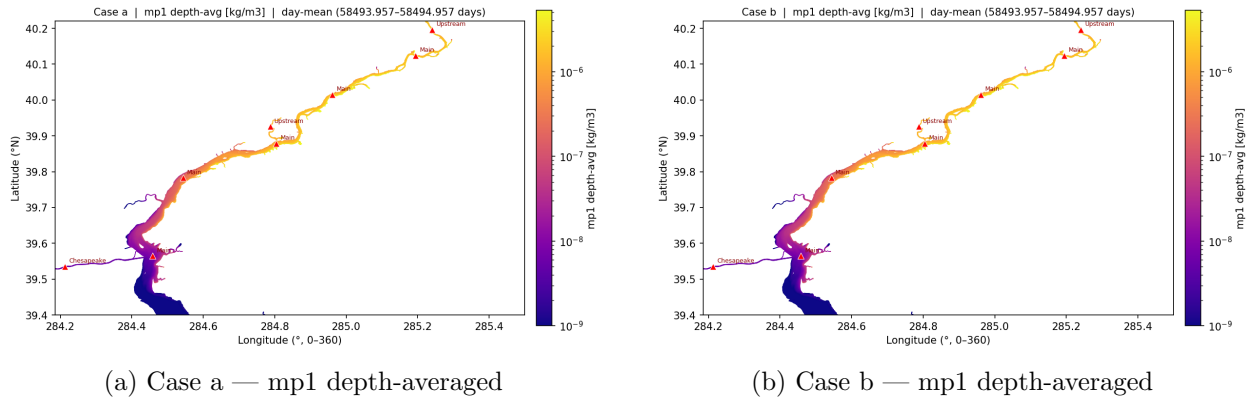


Figure 1: Day-mean depth-averaged mp1 concentration [kg m^{-3}] over the last 24 hours of simulation. Source locations are at the nine river mouths.

Pointwise-Maximum Maps

Cross-Case Difference Maps

Station Time Series

Observations

This section will be completed after reviewing the plots generated by the Python analysis script. The following items should be addressed:

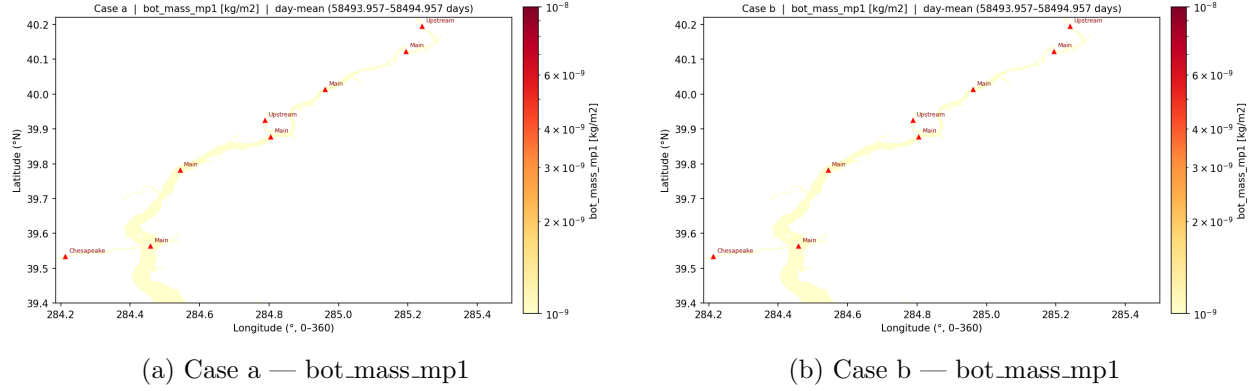


Figure 2: Day-mean bed plastic mass density [kg m^{-2}] over the last 24 hours.

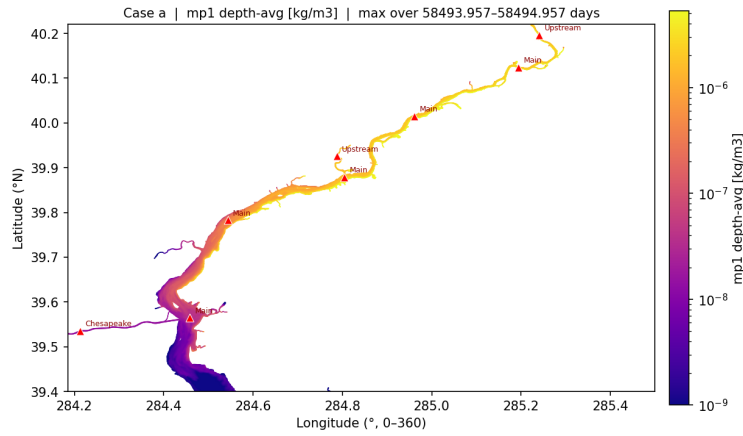


Figure 3: Pointwise maximum depth-averaged mp1 over the last 24 hours, case a. Hotspots (if any) would indicate localised numerical accumulation.

- (1) **Spatial extent of plastic:** does the plume extend realistically downstream of the river sources? Are there any cells with unexpectedly high concentrations ($> 10^{-2} \text{ kg m}^{-3}$)?
- (2) **Bed accumulation pattern:** are bed deposits concentrated near the river mouths and along the main channel, as expected from the tidal forcing?
- (3) **Floc-case differences (cases b/c vs. a):** does aggregation (PLAST_FLOC=T) produce noticeably different horizontal distributions? Where is the sign of the difference most pronounced?
- (4) **Sediment distribution (cases b/c):** is the `coarse_sand_1` field physically consistent with the bathymetry and tidal currents?
- (5) **Station time series:** do the stations downstream of sources show coherent tidal oscillations? Are there stations where the three cases diverge significantly after day 5?
- (6) **Numerical anomalies:** are there isolated hotspot nodes in the pointwise-max maps that are not connected to any river source? If so, report the node index, lon/lat, depth, and peak concentration.

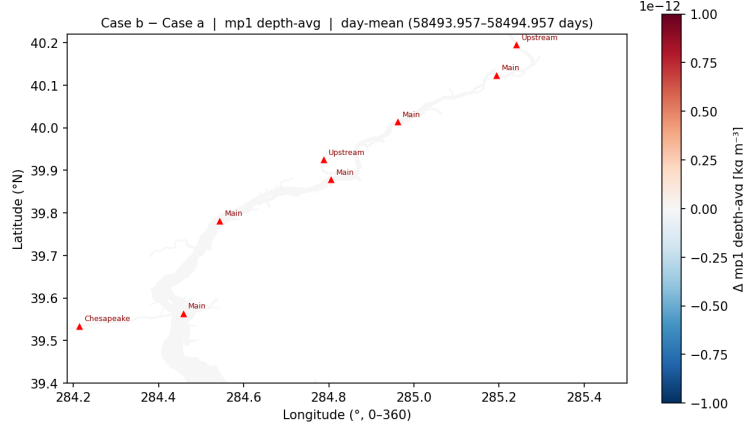


Figure 4: Day-mean mp1 depth-averaged difference: case b – case a. Blue = less plastic in flocculation case; red = more.

Summary and Next Steps

Summary to be written after observations are recorded.

Potential next steps:

- Extended 30-day runs (confirmed to be mass-conservative per Memo 04) to verify longer-term spatial spreading and flocculation-component evolution.
- Open-boundary hypothesis testing: instrument OBC fluxes at the southern bay mouth to quantify tidal export of plastic.
- Fix D: multi-species flux limiting when `SOURCE_AG_RATE` > 0 causes `conc_a` > 0, which is excluded from the current 10-day tests but required for full-physics runs.

Station time series — mp1 depth-averaged [kg m^{-3}]

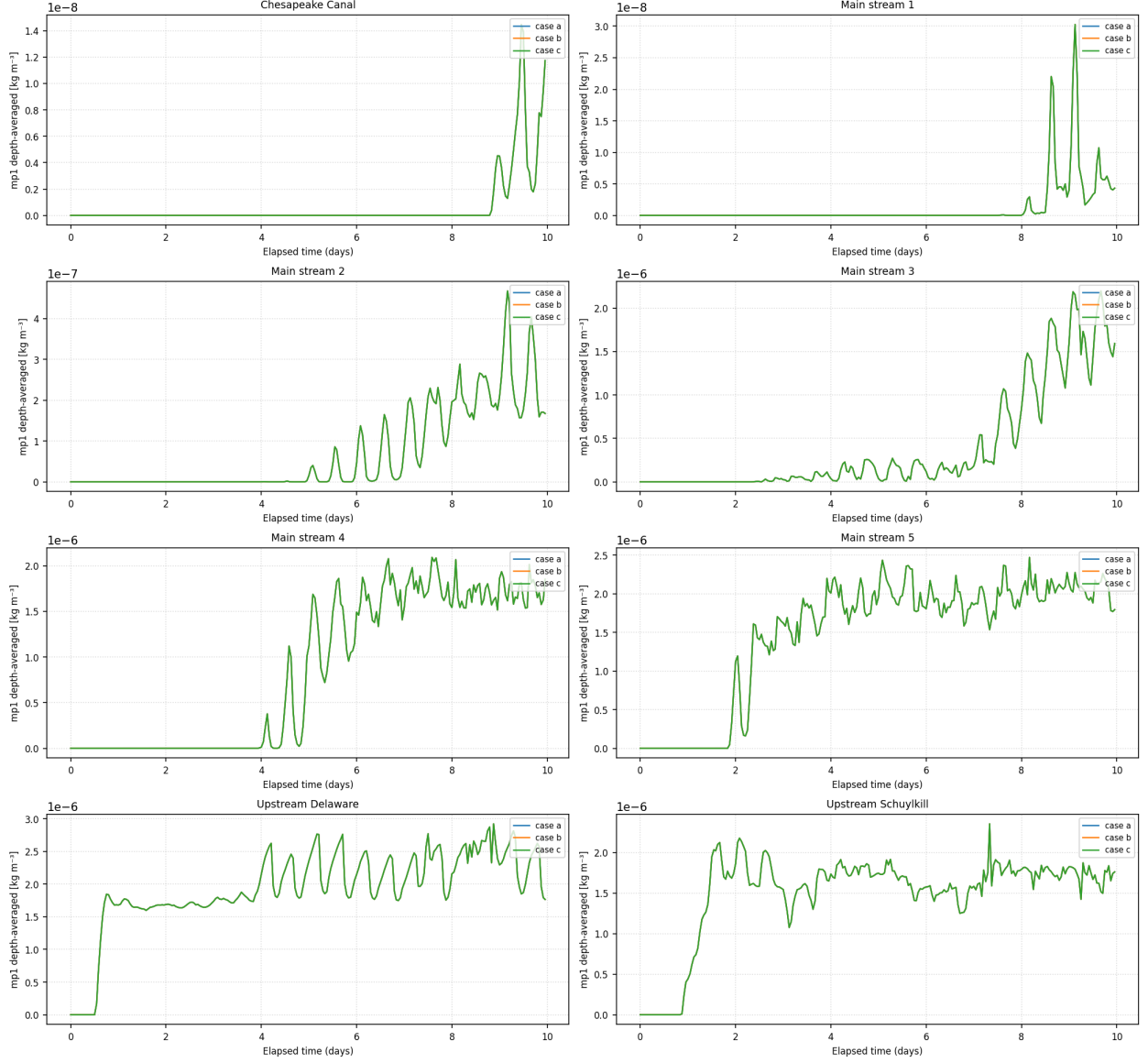


Figure 5: Depth-averaged mp1 time series at the eight monitoring stations for all three cases. Blue: case a; orange: case b; green: case c.